Length of string



```
/*****************************************************************************

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*****************************************************************************/
#include <stdio.h>

int main() {
    char str[100];
    int i = 0;

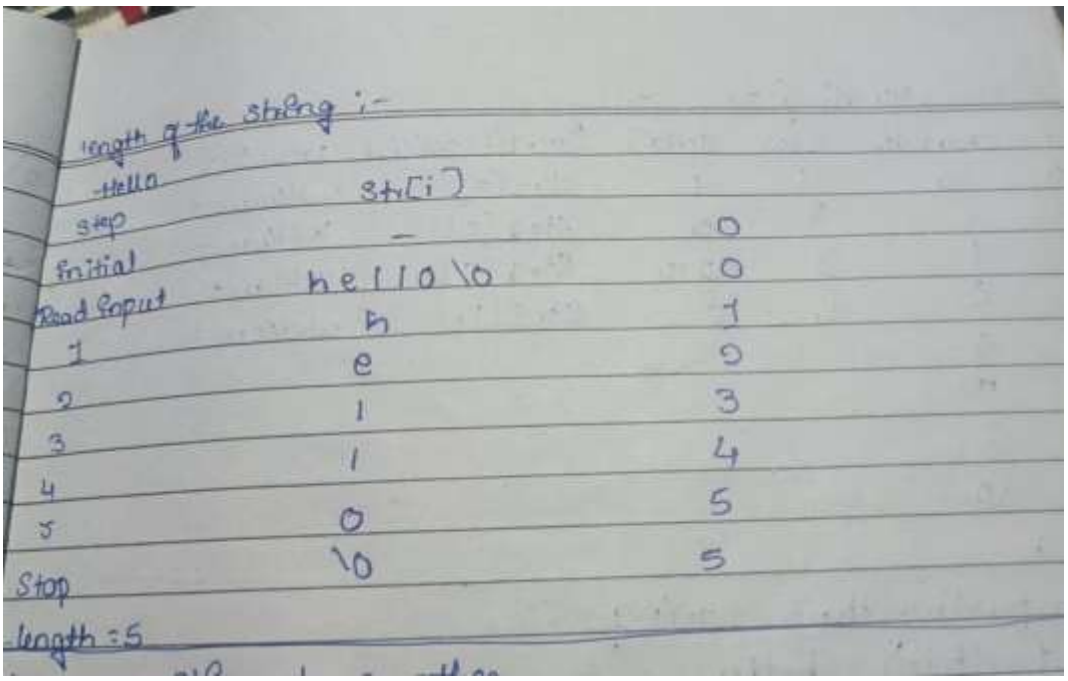
    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);
    while (str[i] != '\0')
        i++;

    printf("Length = %d", i);

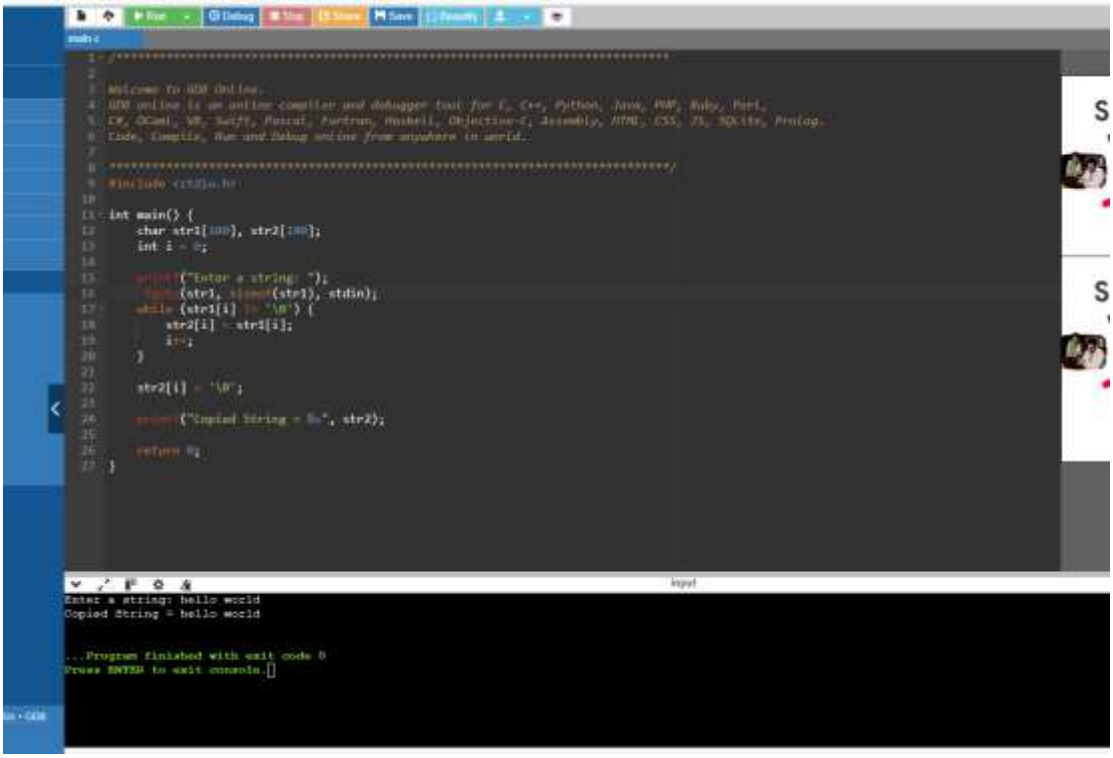
    return 0;
}
```

```
Enter a string: hello
Length = 6

...Program finished with exit code 0
Press ENTER to exit console.
```

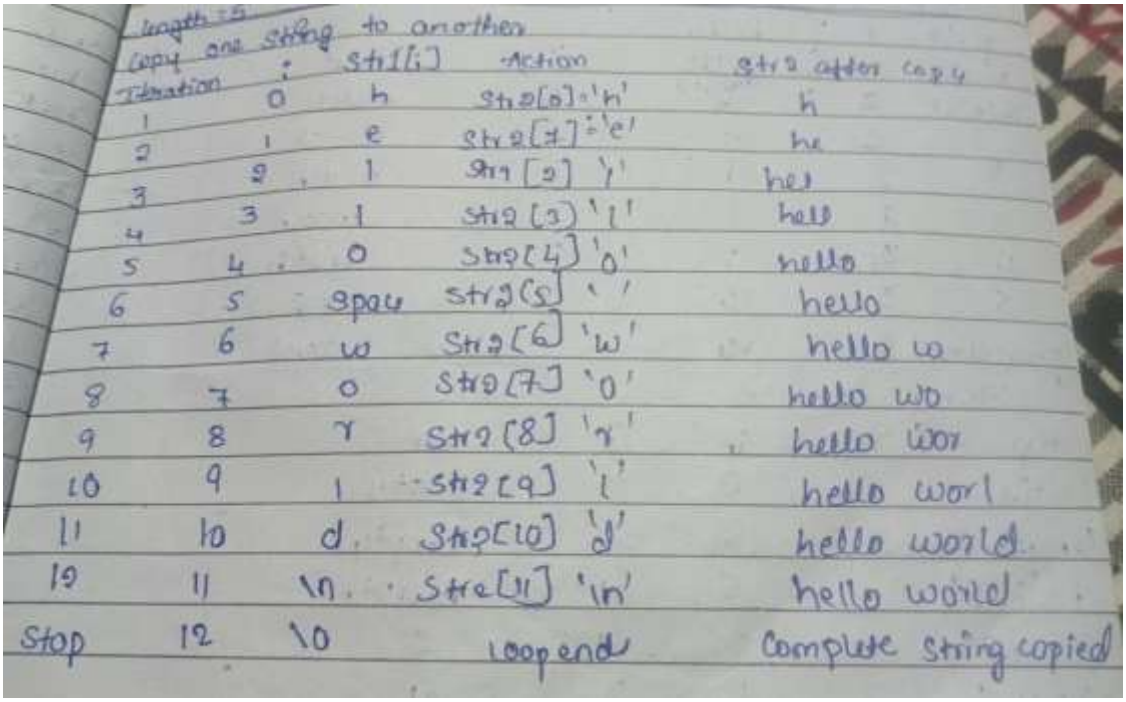


Copy one string to another



```
Enter a string: hello world
Copied String = hello world


...Program finished with exit code 0
Press ENTER to exit console.
```

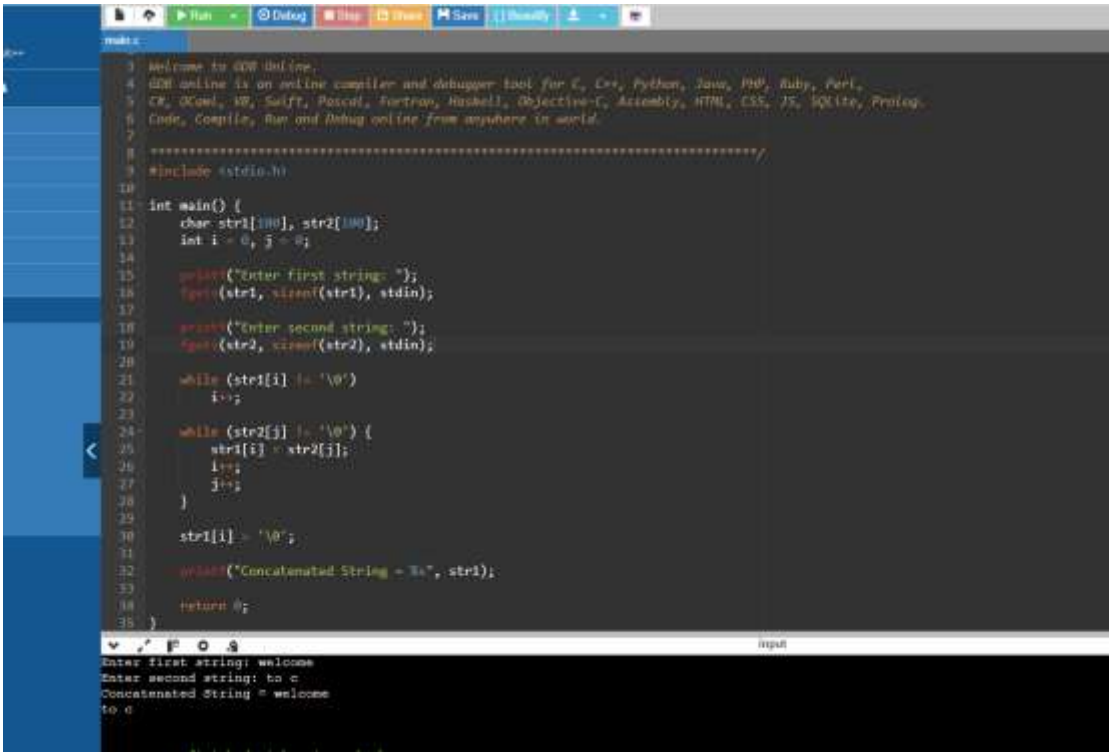


Length = 5

Copy one string to another

Iteration	i	str1[i]	Action	str2 after copy
1	0	h	str2[0] = 'h'	h
2	1	e	str2[1] = 'e'	he
3	2	l	str2[2] = 'l'	hel
4	3	l	str2[3] = 'l'	hell
5	4	o	str2[4] = 'o'	hello
6	5	space	str2[5] = ' '	hello
7	6	w	str2[6] = 'w'	hello w
8	7	o	str2[7] = 'o'	hello wo
9	8	r	str2[8] = 'r'	hello wor
10	9	l	str2[9] = 'l'	hello worl
11	10	d	str2[10] = 'd'	hello world
12	11	\n	str2[11] = '\n'	hello world
Stop	12	\0	loop end	Complete string copied

Concatenate two string



```
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*******************************************************************************/
#include <stdio.h>

int main() {
    char str1[100], str2[100];
    int i = 0, j = 0;

    printf("Enter first string: ");
    fgets(str1, sizeof(str1), stdin);

    printf("Enter second string: ");
    fgets(str2, sizeof(str2), stdin);

    while (str1[i] != '\0')
        i++;

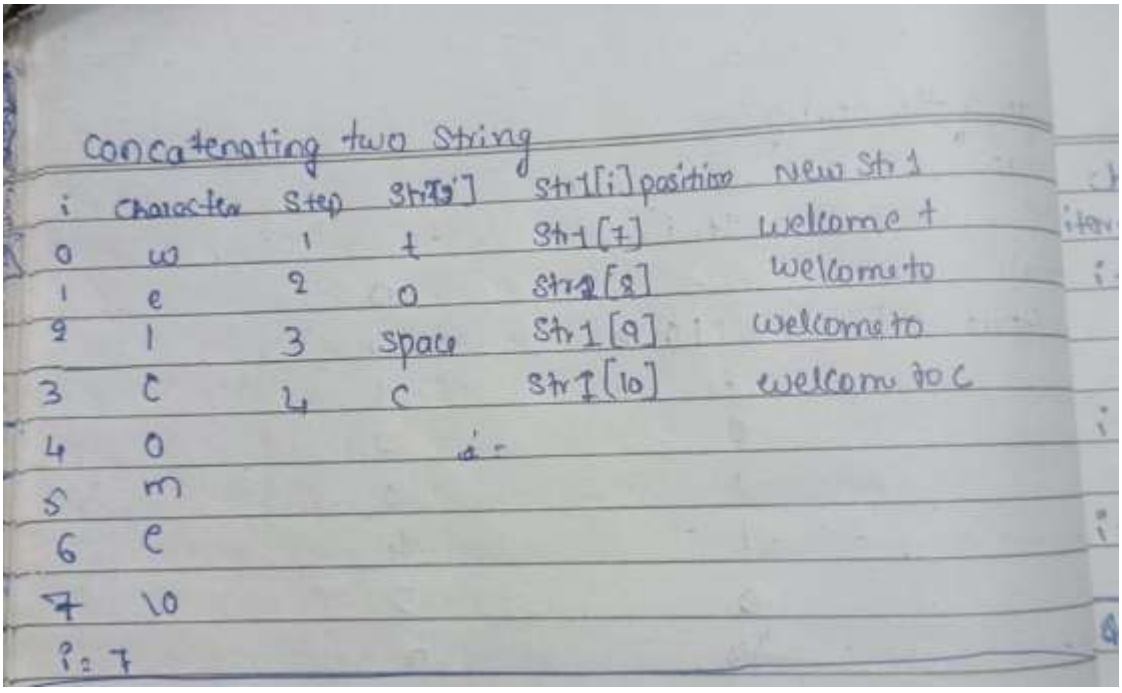
    while (str2[j] != '\0') {
        str1[i] = str2[j];
        i++;
        j++;
    }

    str1[i] = '\0';

    printf("Concatenated String = %s", str1);

    return 0;
}
```

```
Enter first string: welcome
Enter second string: to c
Concatenated String = welcome
to c
```



Concatenating two String

i	character	Step	str2[i]	str1[i] position	New str1
0	w	1	t	str1[7]	welcome t
1	e	2	o	str1[8]	welcome to
2	l	3	space	str1[9]	welcome to
3	c	4	c	str1[10]	welcome to c
4	o				
5	m				
6	e				
7	\0				

i = 7

Comparing two string

```c
/******************************************************************************

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*******************************************************************************/
#include <stdio.h>

int main() {
    char str1[100], str2[100];
    int i = 0;

    printf("Enter first string: ");
    fgets(str1, sizeof(str1), stdin);

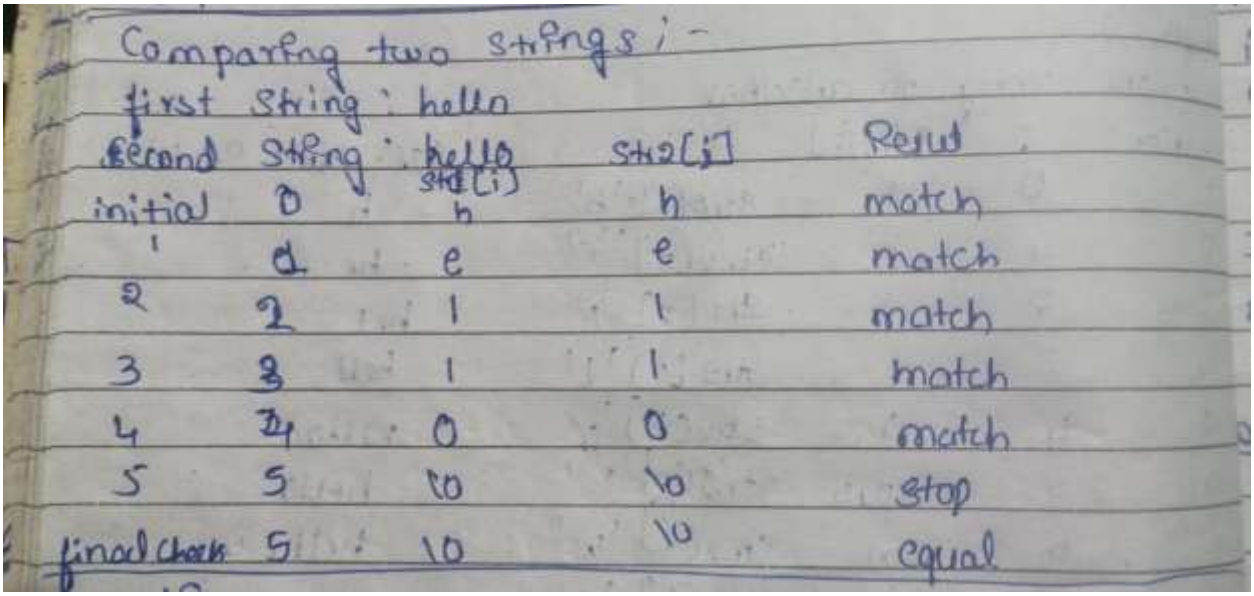

    printf("Enter second string: ");
fgets(str2, sizeof(str2), stdin);
    while (str1[i] == str2[i] && str1[i] != '\0')
        i++;

    if (str1[i] == str2[i])
        printf("Strings are Equal");
    else
        printf("Strings are Not Equal");

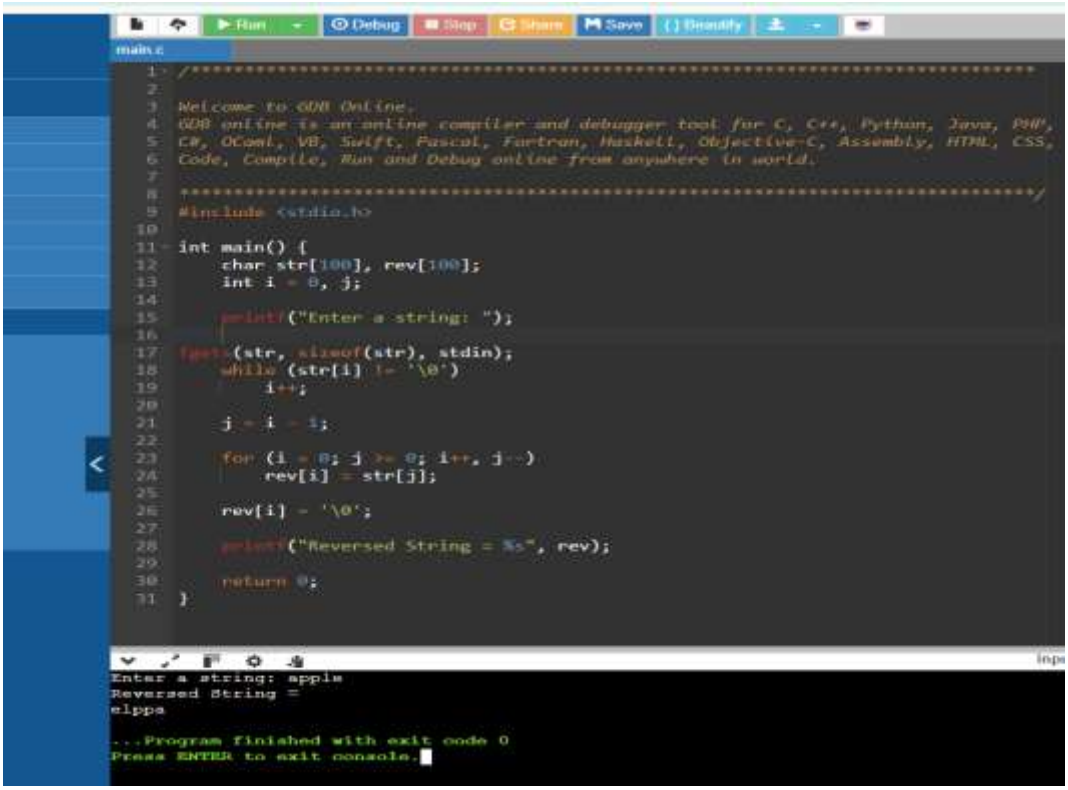
    return 0;
}
```

```
Enter first string: hello
Enter second string: hello
Strings are Equal

...Program finished with exit code 0
Press ENTER to exit console.
```



String reverse



```
/*****************************************************************************

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*****************************************************************************/
#include <stdio.h>

int main() {
    char str[100], rev[100];
    int i = 0, j;

    printf("Enter a string: ");

    fgets(str, sizeof(str), stdin);
    while (str[i] != '\0')
        i++;

    j = i - 1;

    for (i = 0; j >= 0; i++, j--)
        rev[i] = str[j];

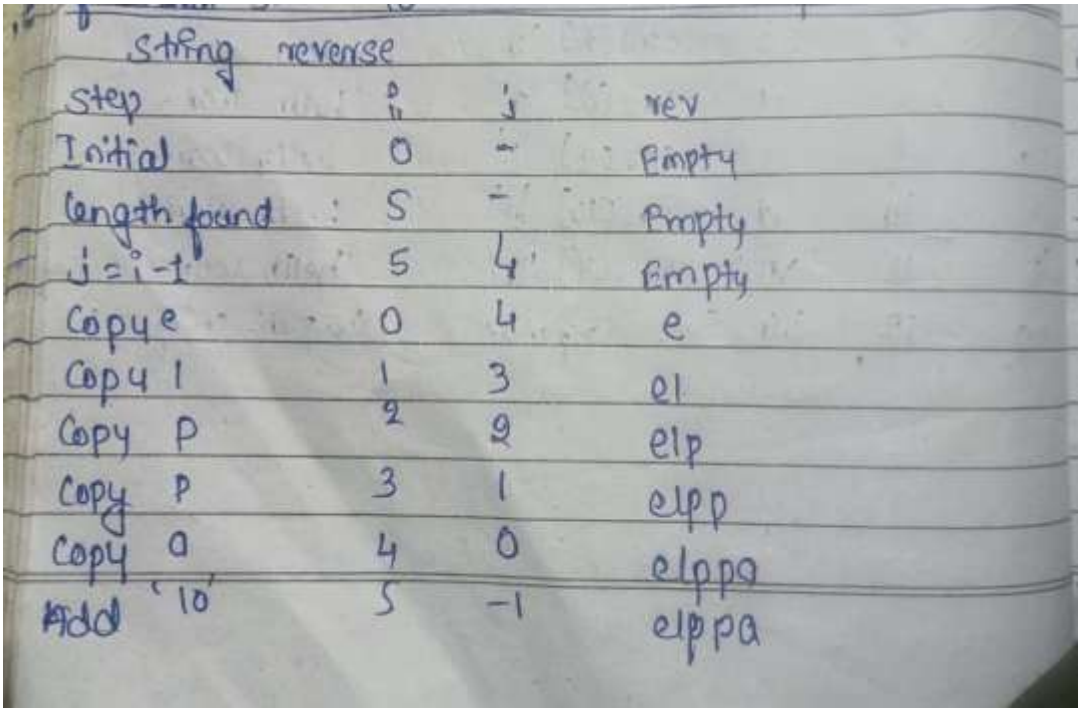
    rev[i] = '\0';

    printf("Reversed String = %s", rev);

    return 0;
}
```

```
Enter a string: apple
Reversed String =
elppa

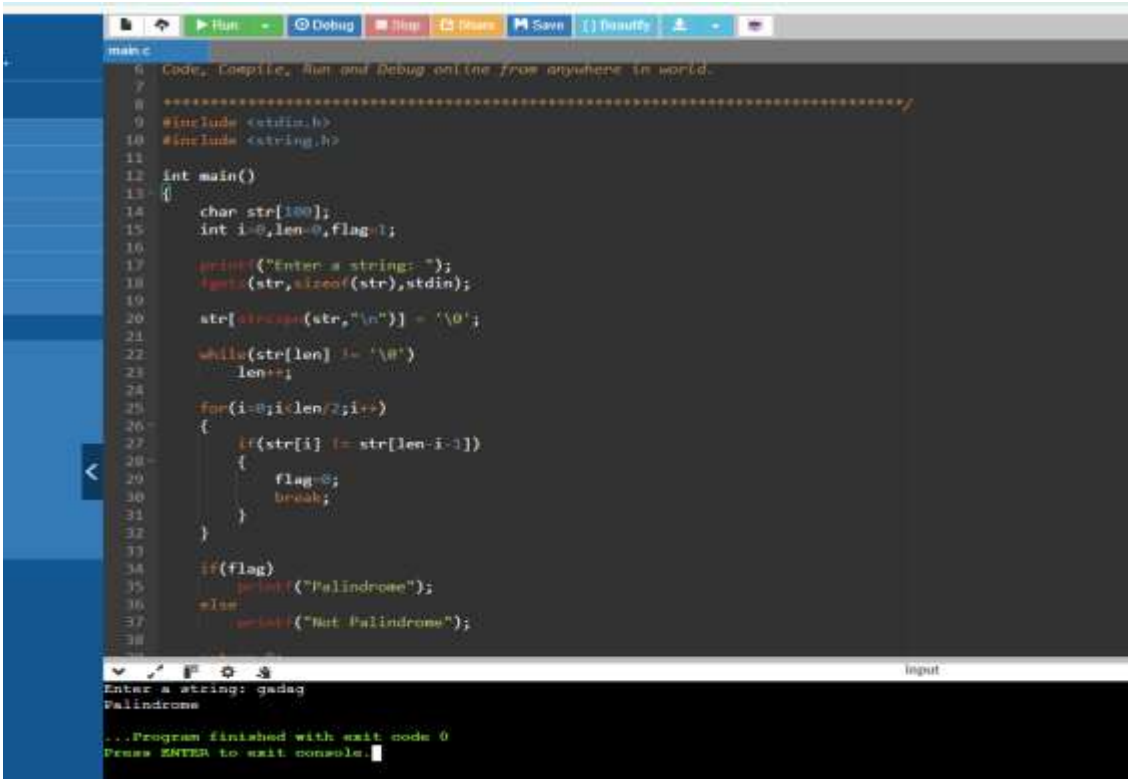
...Program finished with exit code 0
Press ENTER to exit console.
```



String reverse

Step	i	j	rev
Initial	0	-	Empty
Length found	5	-	Empty
j = i - 1	5	4	Empty
Copy e	0	4	e
Copy l	1	3	el
Copy p	2	2	elp
Copy p	3	1	elpp
Copy a	4	0	elppa
Add '\0'	5	-1	elppa

Check palindrome



```
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*******************************************************************************/
#include <stdio.h>
#include <string.h>

int main()
{
    char str[100];
    int i=0,len=0,flag=1;

    printf("Enter a string: ");
    fgets(str,sizeof(str),stdin);

    str[strcspn(str,"\n")] = '\0';

    while(str[len] != '\0')
        len++;

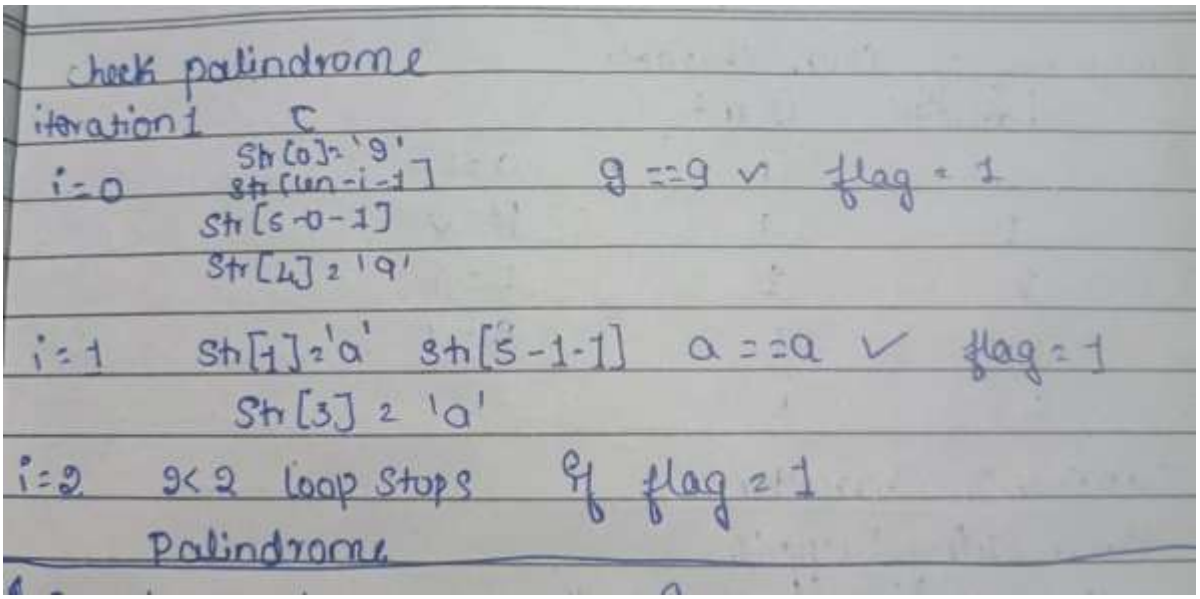
    for(i=0;i<len/2;i++)
    {
        if(str[i] != str[len-i-1])
        {
            flag=0;
            break;
        }
    }

    if(flag)
        printf("Palindrome");
    else
        printf("Not Palindrome");

}
```

```
Input
Enter a string: gadag
Palindrome

...Program finished with exit code 0
Press ENTER to exit console.
```



check palindrome

iteration 1 C

i=0 Str[0]='g' Str[len-i-1] g==g ✓ flag = 1

Str[5-0-1]

Str[4] = 'g'

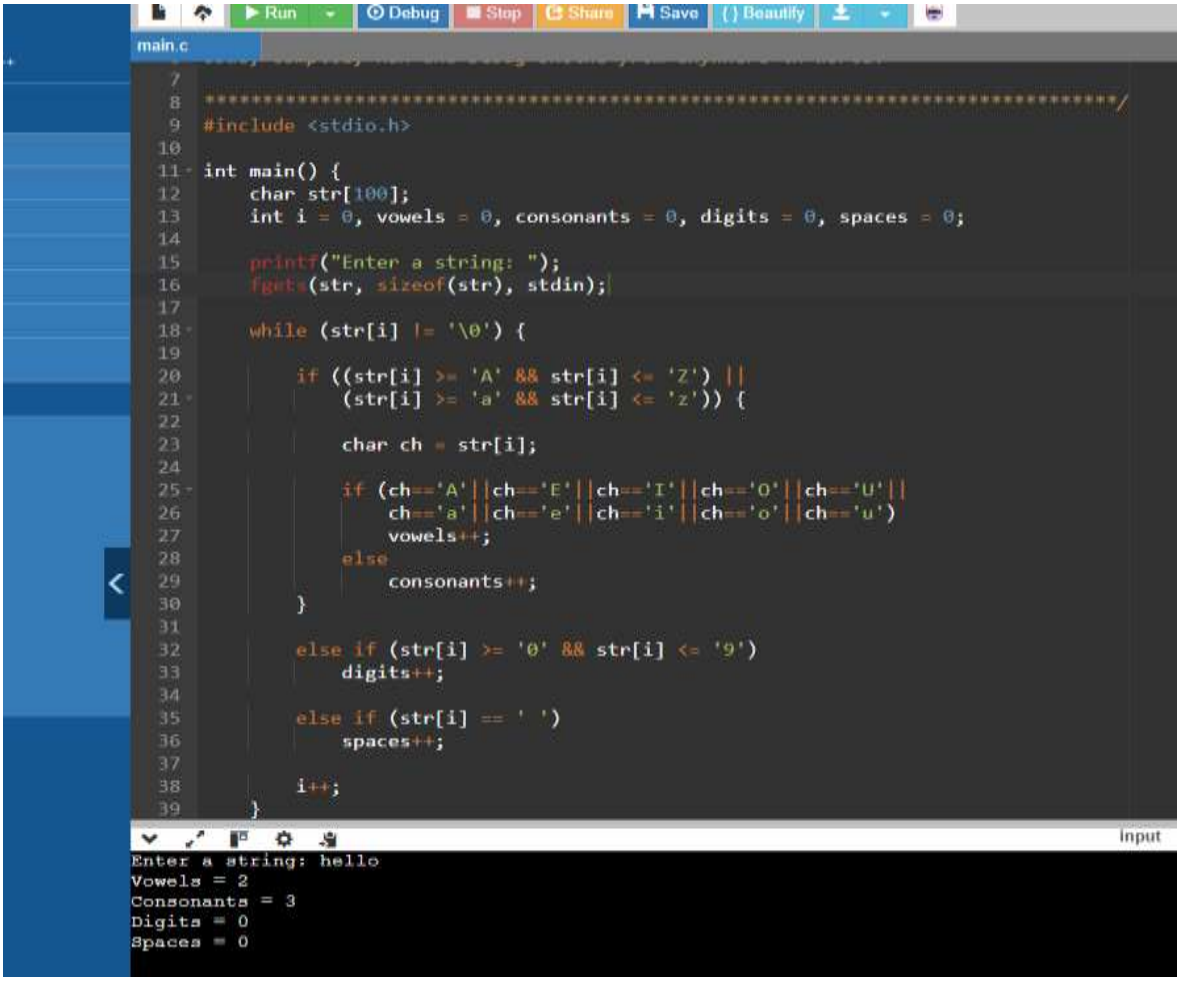
i=1 Str[1]='a' Str[5-1-1] a==a ✓ flag=1

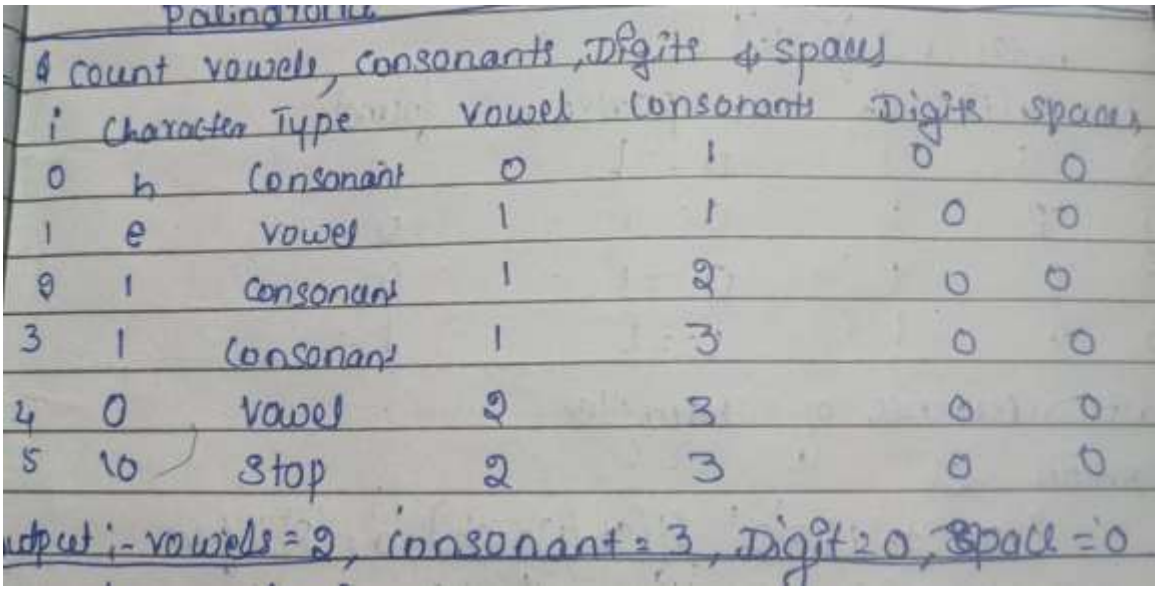
Str[3] = 'a'

i=2 2<2 loop stops & flag=1

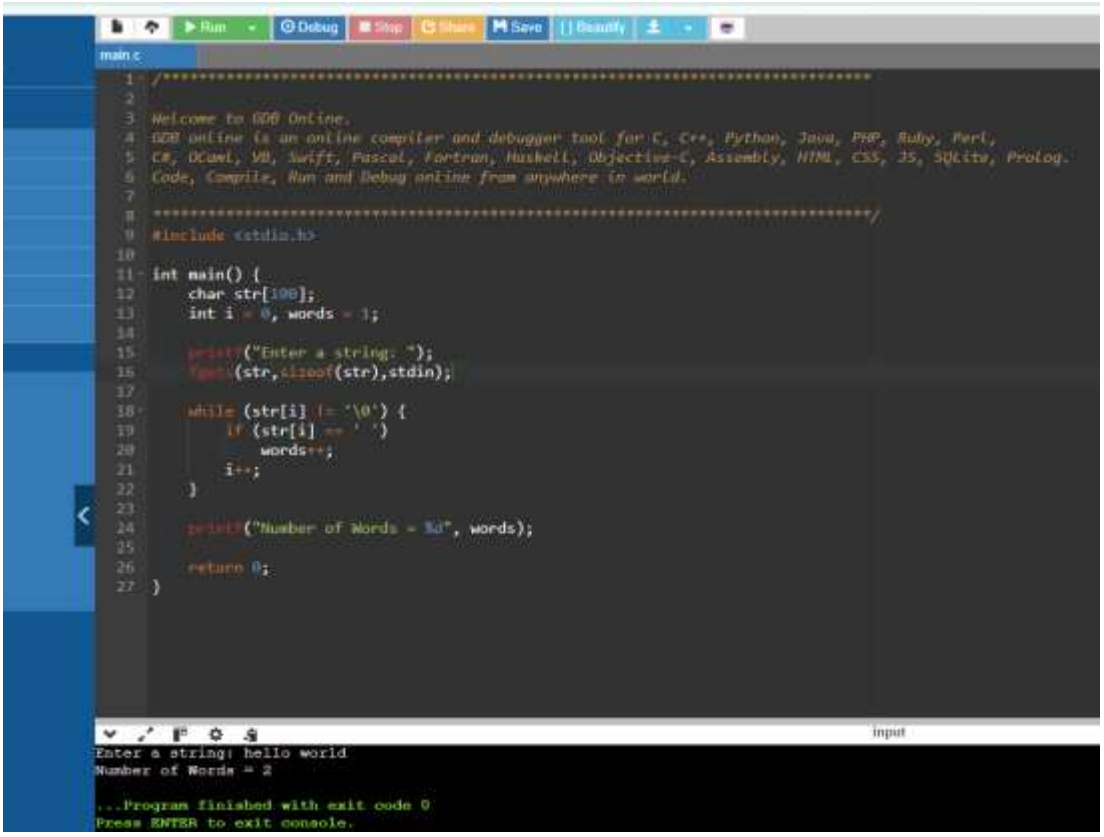
Palindrome

Count vowels consonants





Count words in string



```
/******************************************************************************

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*******************************************************************************/
#include <stdio.h>

int main() {
    char str[100];
    int i = 0, words = 1;

    printf("Enter a string: ");
    fgets(str,sizeof(str),stdin);

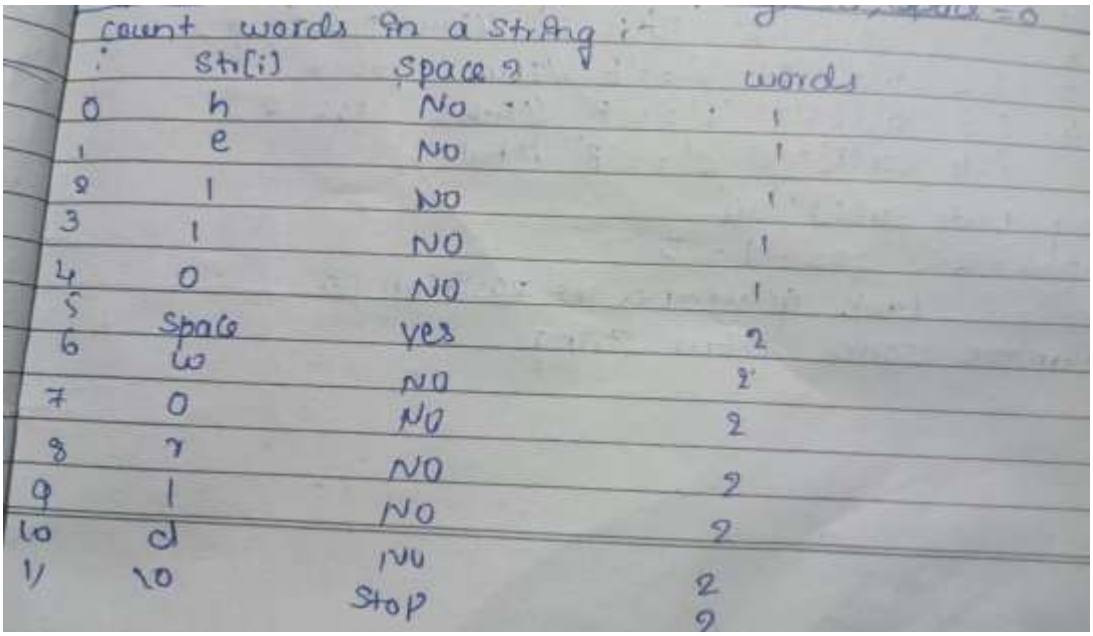
    while (str[i] != '\0') {
        if (str[i] == ' ')
            words++;
        i++;
    }

    printf("Number of Words = %d", words);

    return 0;
}
```

```
Enter a string: hello world
Number of Words = 2

...Program finished with exit code 0
Press ENTER to exit console.
```



count words in a string :-

	str[i]	space ?	words
0	h	No	1
1	e	No	1
2	l	No	1
3	l	No	1
4	o	No	1
5	space	yes	2
6	w	No	2
7	o	No	2
8	r	No	2
9	l	No	2
10	d	No	2
11	\0	Stop	2

Frequency of each character

```c
/*****************************************************************************
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*****************************************************************************/
#include <stdio.h>
#include<string.h>
int main() {
    char str[100];
    int i, j, count;

    printf("Enter a string: ");
fgets(str, sizeof(str), stdin);
str[strcspn(str, "\n")] = '\0';
    for (i = 0; str[i] != '\0'; i++) {

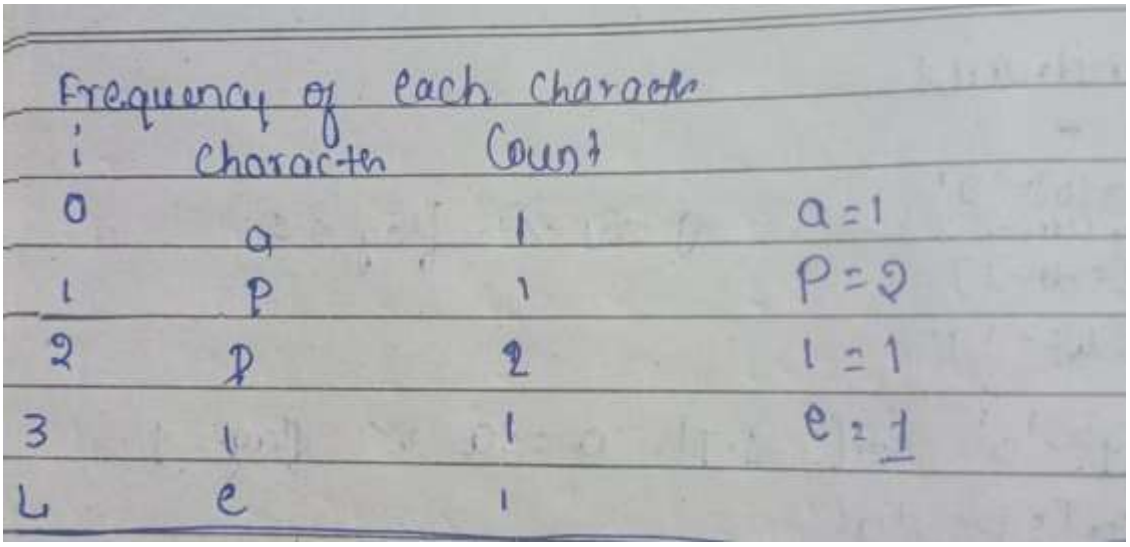
        count = 1;

        if (str[i] == '*')
            continue;

        for (j = i + 1; str[j] != '\0'; j++) {
            if (str[i] == str[j]) {
                count++;
                str[j] = '*';
            }
        }

        printf("%c = %d\n", str[i], count);
    }
```

```
Input
Enter a string: apple
a = 1
p = 2
l = 1
e = 1
```



Occurrence of characte

```c
/*
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*******************************************************************************/
#include <stdio.h>
#include <string.h>

int main() {
    char str[100], ch;
    int i, found = 0;

    printf("Enter a string: ");
    fgets(str, sizeof(str), stdin);
    str[strcspn(str, "\n")] = '\0';

    printf("Enter character: ");
    scanf("%c", &ch);

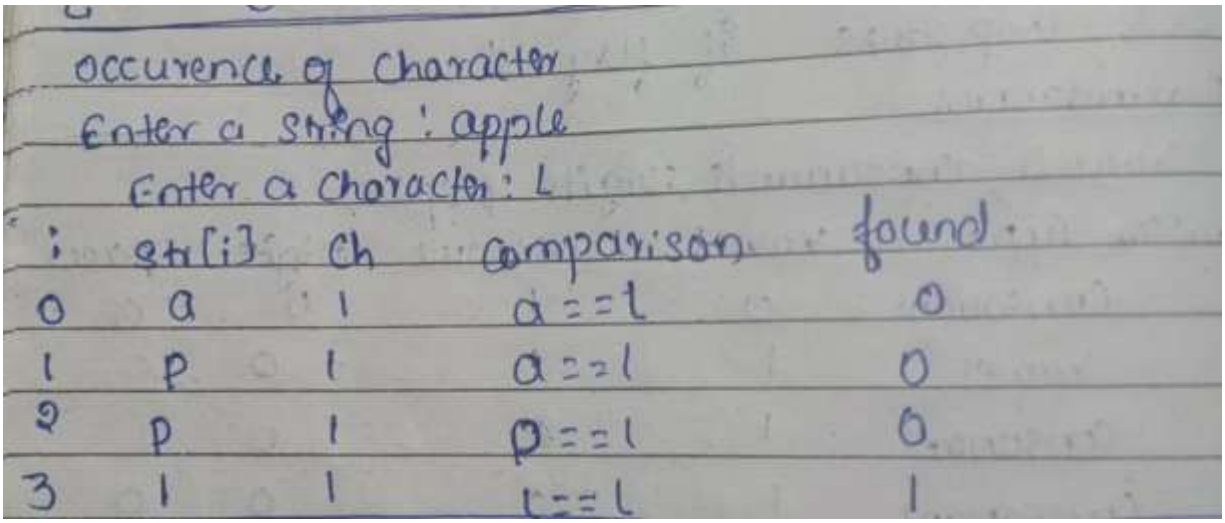
    for(i = 0; str[i] != '\0'; i++) {
        if(str[i] == ch) {
            printf("First occurrence at position %d", i + 1);
            found = 1;
            break;
        }
    }

    if(!found)
        printf("Character not found");

    return 0;
}
```

```
Input
Enter a string: apple
Enter character: l
First occurrence at position 4

...Program finished with exit code 0
Press ENTER to exit console.
```



Last occurance of characterr

```c
/*****************************************************************************

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*****************************************************************************/
#include <stdio.h>
#include <string.h>

int main() {
    char str[100], ch;
    int i, pos = -1;

    printf("Enter string: ");
    fgets(str, sizeof(str), stdin);
    str[strcspn(str, "\n")] = '\0';

    printf("Enter character: ");
    scanf("%c", &ch);

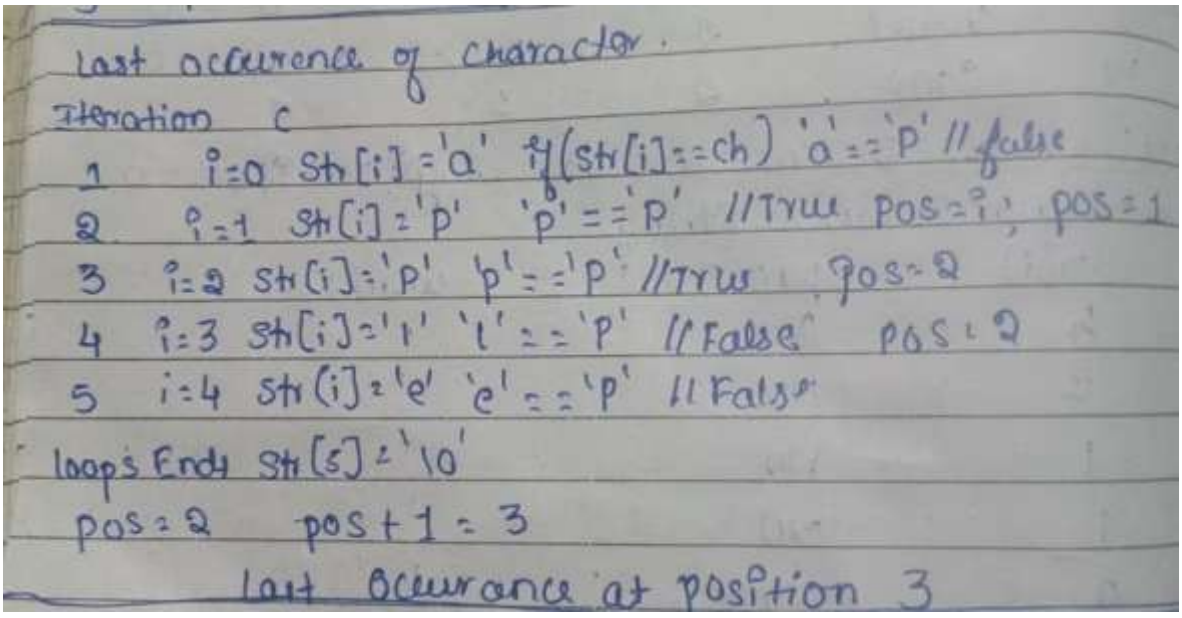
    for(i = 0; str[i] != '\0'; i++)
        if(str[i] == ch)
            pos = i;

    if(pos != -1)
        printf("Last occurrence at position %d", pos + 1);
    else
        printf("Character not found");

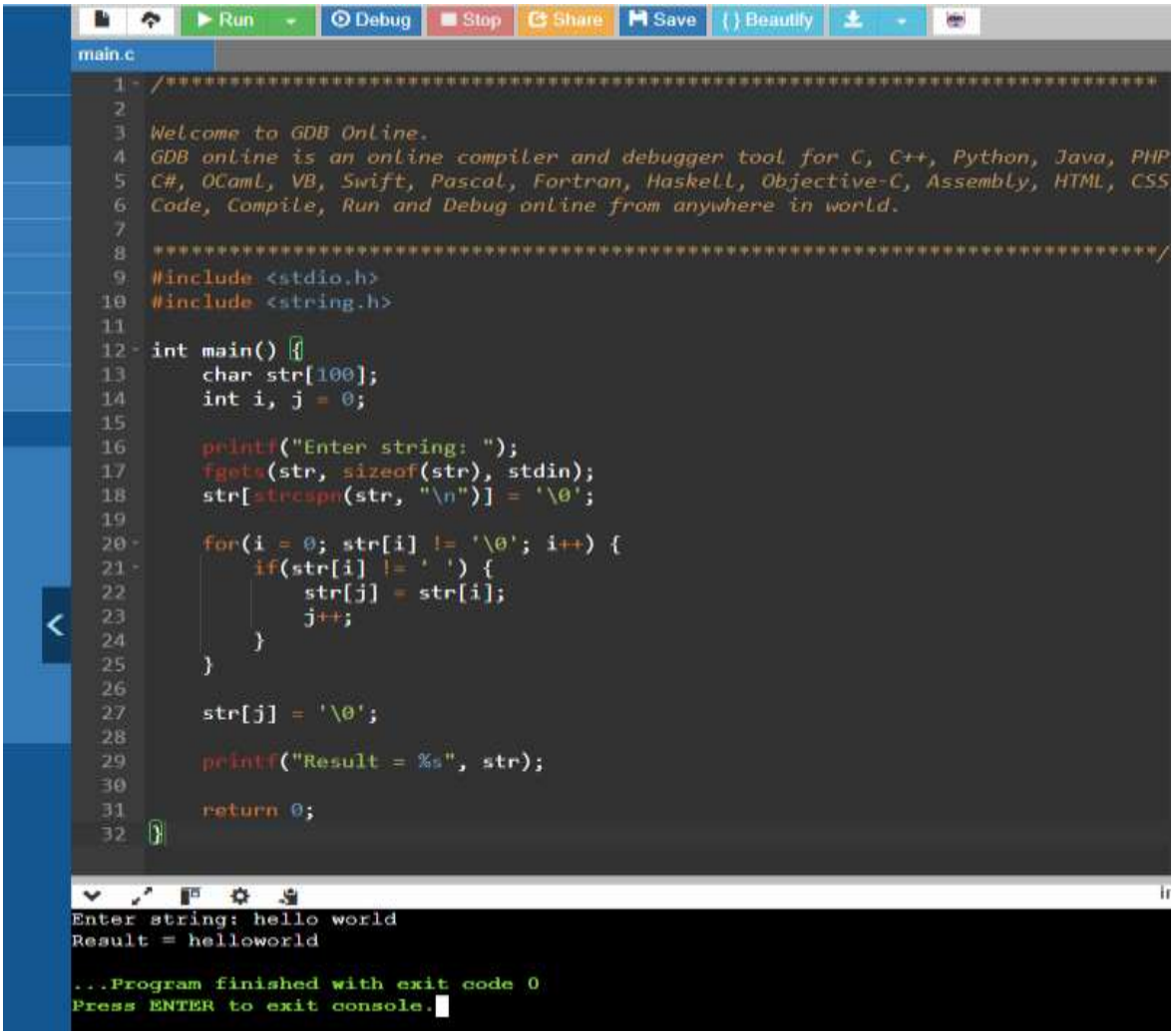
    return 0;
}
```

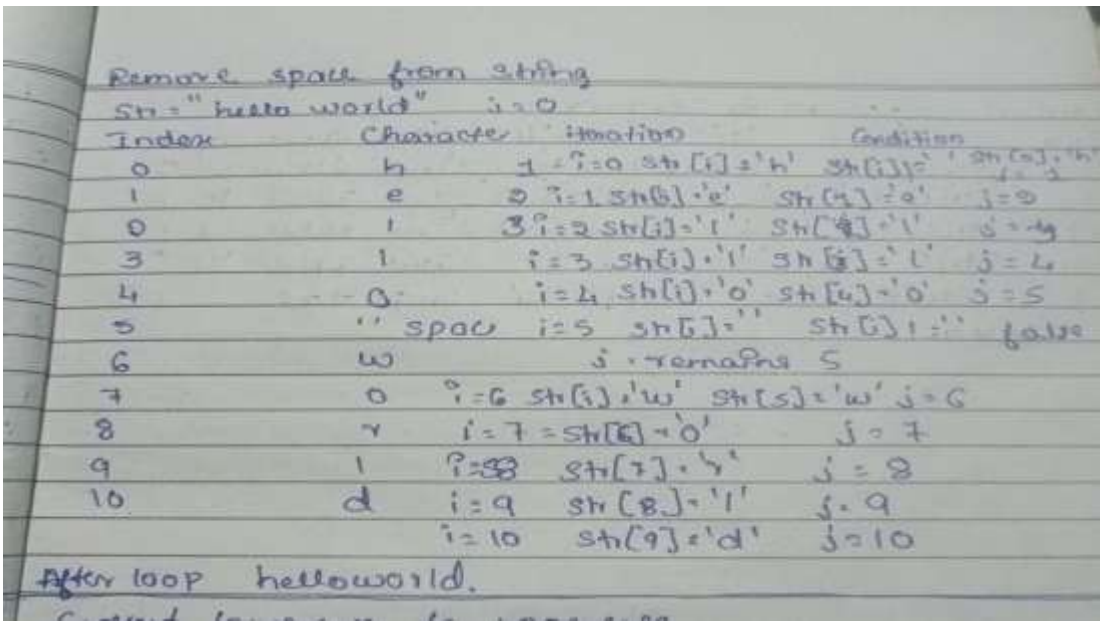
```
Enter string: apple
Enter character: p
Last occurrence at position 3

...Program finished with exit code 0
Press ENTER to exit console.
```



Remove space from string





Convert lower case to upper case

```c
/******************************************************************************

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*******************************************************************************/
#include <stdio.h>
#include <string.h>

int main() {
    char str[100];
    int i;

    printf("Enter string: ");
    fgets(str, sizeof(str), stdin);
    str[strcspn(str, "\n")] = '\0';

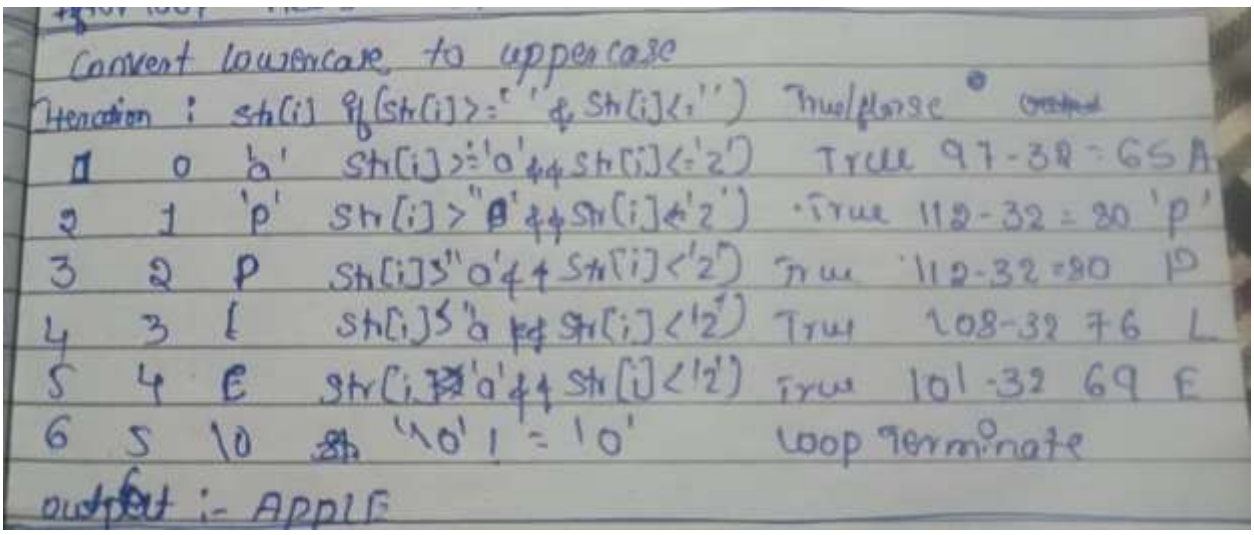
    for(i = 0; str[i] != '\0'; i++) {
        if(str[i] >= 'a' && str[i] <= 'z')
            str[i] = str[i] - 32;
    }

    printf("%s", str);

    return 0;
}
```

```
Enter string: apple
APPLE

...Program finished with exit code 0
Press ENTER to exit console.
```



Convert upper case to lower case

```c
/*****************************************************************************

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*****************************************************************************/
#include <stdio.h>
#include <string.h>

int main() {
    char str[100];
    int i;

    printf("Enter string: ");
    fgets(str, sizeof(str), stdin);
    str[strcspn(str, "\n")] = '\0';

    for(i = 0; str[i] != '\0'; i++) {
        if(str[i] >= 'A' && str[i] <= 'Z')
            str[i] = str[i] + 32;
    }

    printf("%s", str);

    return 0;
}
```

```
Enter string: HELLO
hello

...Program finished with exit code 0
Press ENTER to exit console.
```

